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Greedy Solutions

Solution 1:

Time Complexity : o(n)
Space Complexity: o(1)

```
import java.util.*;

class Solution {

static int BalancedPartition(String str, int n){

        if (n == 0)
                return 0;

        int r = 0, l = 0;
        int ans = 0;
        for(int i = 0; i < n; i++)  {
        if (str.charAt(i) == 'R'){
                r++;
        }

        else if (str.charAt(i) == 'L'){
                l++;
        }

        if (r == l){
                ans++;
        }
        }

        return ans;
}

public static void main(String[] args){
        String str = "LLRRRLLRRL";
        int n = str.length();

        System.out.print(BalancedPartition(str, n) + "\n");
}
}
```

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Solution 2 :

Time Complexity : o(1)
Space Complexity: o(1)

```
class Solution {

	public static int kthOdd(int[] range, int K) {

		if (K <= 0)
			return 0;

		int L = range[0];
		int R = range[1];

		if ((R & 1) > 0) {
			int Count = (int) Math.ceil((R - L + 1) / 2);
			if (K > Count)
				return 0;
			else
				return (R - 2 * K + 2);
		} else {
			int Count = (R - L + 1) / 2;
			if (K > Count)
				return 0;
			else
				return (R - 2 * K + 1);
		}
	}

	public static void main(String args[]){
		int[] p = { -10, 10 };
		int k = 8;
		System.out.println(kthOdd(p, k));
	}
}
```

Solution 3 :

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Time Complexity : o(n)
Space Complexity: o(n)

```
import java.util.Arrays;
public class Main {
	public static char[] lexo_small(int n, int k){
		char arr[] = new char[n];
		Arrays.fill(arr, 'a');
		for (int i = n - 1; i >= 0; i--) {
			k -= i;
			if (k >= 0) {
				if (k >= 26) {
					arr[i] = 'z';
					k -= 26;
				}

				else {
					arr[i] = (char)(k + 97 - 1);
					k -= arr[i] - 'a' + 1;
				}
			}
			else
				break;

			k += i;
		}
		return arr;
	}

	public static void main(String[] args){
		int n = 5, k = 42;

		char arr[] = lexo_small(n, k);

		System.out.print(new String(arr));
	}
}
```

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Solution 4 :

Time Complexity : o(n)
Space Complexity: o(1)

```
class Solution {
static int maxProfit(int prices[], int n)
{
        int buy = prices[0], max_profit = 0;
        for (int i = 1; i < n; i++) {
        if (buy > prices[i])
                buy = prices[i];
        else if (prices[i] - buy > max_profit)
                max_profit = prices[i] - buy;
        }
        return max_profit;
}

public static void main(String args[]){
        int prices[] = { 7, 1, 5, 6, 4 };
        int n = prices.length;
        int max_profit = maxProfit(prices, n);
        System.out.println(max_profit);
}
}
```

Solution 5 :

Time Complexity :O((N−1)c(K−1))
(Here 'c' here depicts combinations i.e. ((n-1)!/((n-k)!*(k-1)!) Where N is the number of elements of the array and K is the number of divisions that we are having)
Space Complexity: o(n)

```
class Solution {
        public static int ans = 10000000;
        public static void solve(int a[], int n, int k,
                                                int index, int sum, int maxsum){
```

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```
            if (k == 1) {
                maxsum = Math.max(maxsum, sum);
                sum = 0;
                for (int i = index; i < n; i++) {
                    sum += a[i];
                }
                maxsum = Math.max(maxsum, sum);
                ans = Math.min(ans, maxsum);
                return;
            }
            sum = 0;
            for (int i = index; i < n; i++) {
                sum += a[i];
                maxsum = Math.max(maxsum, sum);
                solve(a, n, k - 1, i + 1, sum, maxsum);
            }
        }
        public static void main(String[] args){
            int arr[] = { 1, 2, 3, 4 };
            int k = 3; // K divisions
            int n = 4; // Size of Array
            solve(arr, n, k, 0, 0, 0);
            System.out.println(ans + "\n");
        }
}
```

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